

Mohammad Farid Azampour

POSTDOCTORAL RESEARCHER

IFL - Interdisziplinäres Forschungslabor, Klinikum rechts der Isar, Ismaninger Str. 22, 81675 München
Technical University of Munich

Chair of Computer Aided Medical Procedures & Augmented Reality (CAMP)

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Professional Employment

Postdoctoral Researcher

TECHNICAL UNIVERSITY OF MUNICH (TUM)

- Chair for Computer Aided Medical Procedures with Nassir Navab.

Munich, Germany

09/2024 – Present

Research Fellow

TECHNICAL UNIVERSITY OF MUNICH (TUM)

- Chair for Computer Aided Medical Procedures with Nassir Navab.

Munich, Germany

02/2021 – 09/2024

Visiting Researcher

IMPERIAL COLLEGE LONDON

- Biomedical Image Analysis group in the Department of Computing with Bernhard Kainz.

London, UK

10/2020 – 04/2021

Technical Director

DARIC PARS SOLUTIONS

- Designed a tourism wallet for areas with limited access to major payment services.

Tehran, Iran

06/2016 – 02/2019

Data Scientist

TARAABARNET

- Examined truck–goods relationships to extract patterns and propose a pricing algorithm using 40 million data points.

Tehran, Iran

04/2018 – 06/2018

Software Developer | R&D Member

ETICK PARS INTELLIGENT TECHNOLOGIES DEVELOPMENT

- Developed embedded navigation applications using Qt and C# for public transport devices.

Tehran, Iran

09/2011 – 05/2015

Education

PhD in Bio Electrical Engineering

SHARIF UNIVERSITY OF TECHNOLOGY

Thesis: *Multimodal medical image registration using deep learning*

Supervisor: Dr. Emad Fatemizadeh

GPA: **18.60/20** (16 credits)

Tehran, Iran

09/2015 – 09/2024

Graduate Studies in Bio Electrical Engineering

SHARIF UNIVERSITY OF TECHNOLOGY

Thesis: *Image Registration Using Manifold Learning Based Methods*

Supervisor: Dr. Emad Fatemizadeh

GPA: **17.07/20** (32 credits)

Tehran, Iran

09/2011 – 01/2014

B.Sc. in Electrical Engineering

SHARIF UNIVERSITY OF TECHNOLOGY

GPA: **17.7/20** (140 credits)

Tehran, Iran

09/2007 – 09/2011

Service & Leadership

ACADEMIC SERVICE AND TEACHING

- Member of the Coordination Committee, Chair of CAMP 2024 – Present
- Organizer, Practical Course: *Computational Engineering* 2024 – Present
- Organizer, Practical Course: *Project Management and Software Development for Medical Applications* 2023 – 2024
- Lecturer, *CAMP I & II* 2023 – Present

CONFERENCE ORGANIZATION

- (Co-)Organizer, ASMUS Workshop at MICCAI 2026

JOURNAL REVIEWING

- *Medical Physics*
- *IEEE Journal of Biomedical and Health Informatics (JBHI)*
- *IEEE Transactions on Medical Imaging (TMI)*
- *IEEE Robotics and Automation Letters (RA-L)*
- *Medical Image Analysis (MedIA)*

CONFERENCE REVIEWING

- Medical Image Computing and Computer Assisted Interventions (MICCAI)
- International Conference on Information Processing in Computer-Assisted Interventions (IPCAI)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE/CVF International Conference on Computer Vision (ICCV)
- European Conference on Computer Vision (ECCV) *Outstanding Reviewer Award, 2024*
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

Research Support

Secured competitive funding with total and share amounts indicated below.

Global Incentive Fund

Research Exchange ICL-TUM

€10,000

2020

Role: Lead author

PI: Prof. Navab

Global Incentive Fund

Research Exchange Between Ecole Nantes-TUM

€5,000

2022

Role: Coordinator

PI: Prof. Navab

Bayerische Transformations- und Forschungsstiftung

Development of a High-Precision, Real-Time Positioning System for Intracardiac Implant Placement (ENTWICKLUNG EINES HOCHPRÄZISEN UND ECHTZEITFÄHIGEN POSITIONIERUNGSSYSTEMS ZUR INTRAKARDIALEN IMPLANTAT PLATZIERUNG)

€1,000,000 (share: €280,000)

Pending final approval

Role: Lead author & Coordinator

PI: Prof. Navab

Mentorship & Teaching

SUPERVISION

- **PhD Mentor:** Mentor of three PhD students as of Oct. 2023
- **Co-supervisor of master and bachelor theses:** Successfully co-supervised more than 15 M.Sc. and B.Sc. theses

MSc THESIS SUPERVISION

- **Hannes Hase:** Ultrasound-Guided Robotic Navigation with Deep Reinforcement Learning. *Publication: [C-2].*
- **Navneet Madhu Kumar:** Uncertainty-driven Ultrasound Pose Estimation.
- **Kristina Mach:** MR-TRUS deformable registration using deep generative models.
- **Jane Lameski:** Biomechanical Constraint Spine CT-US Registration. *Publication: [J-6].*
- **Magdalena Wysocki:** Neural Radiance Fields for Ultrasound Imaging. *Publication: [A-1].*
- **Miruna Gafencu:** Shape completion in the dark: completing vertebrae morphology from 3D ultrasound. *Publication: [J-5].*
- **Marina Dominguez:** Diffusion as Sound Propagation: Physics-inspired Model for Ultrasound Image Generation. *Publication: [C-17].*
- **Philipp (Jannis) Müller-Hermann:** Supervised X-ray to Angiography Image Translation with Image Inpainting.
- **Manuel Obelleiro Liz:** High-Resolution CTCA Image Synthesis with Patched Semantically Conditioned Diffusion Models.
- **Luis Reyes:** Compact Neural Networks for Efficient Iterative Skeletonization. *Publication: [C-20].*
- **Daniil Morozov:** 3D Cross-modal Keypoint Descriptor for MR-US Matching and Registration.
- **Sami Saidi:** Optimizing Treatment Margin Assessment in Lung Cancer Cryoablation: Advanced 3D CT Registration Techniques.
- **Janluka Janelidze:** Accelerated High-Resolution 3D Medical Image Synthesis with Semantic Conditioning for Coronary CTA. *Publication: [C-24].*
- **Haoyang Sun:** Stepwise Topology Explicitly Preserved (STEP) Inference of Coronary Vessel Segmentation.
- **Max Fehrentz:** CrossPose: Intraoperative Cross-Modal 2D/3D Registration with Foundation Models & 3D Gaussian Splatting. *Publication: [C-15].*
- **Zeineb Haouari:** Aligning Diffusion Priors for Enhanced Domain Generalization in Ultrasound Image Segmentation.
- **Bailiang Jiang:** Weakly-supervised biomechanically-constrained CT/MRI registration of the spine. *Publication: [C-4].*
- **Chiara Rita Piemontese:** Target-Driven Autonomous Robotic Ultrasound Probe Positioning for Anatomies Occluded by Acoustic Shadows. *Publication: IROS 2026 (to be submitted).*
- **Davyd Podolskyi:** Enhancing Intraoperative Registration with Neural Radiance Fields: An Exploration of Loss Functions Effects.
- **Marc Balle:** From Explicit to Implicit Conditioning of Deep Segmentation Networks Using Hypernetworks. *Publication: [C-23].*

PHD PLACEMENTS

- **Kristina Mach:** PhD at Technical University of Munich (TUM)

- **Magdalena Wysocki:** PhD at Technical University of Munich (TUM)
- **Miruna Gafencu:** PhD at Technical University of Munich (TUM)
- **Marina Dominguez:** PhD at Universitat Politècnica de València
- **Manuel Obelleiro Liz:** PhD at Universitat Politècnica de València
- **Luis Reyes:** PhD at Technical University of Munich (TUM)
- **Max Fehrentz:** PhD at Technical University of Munich (TUM)
- **Bailiang Jiang:** PhD at Technical University of Munich (TUM)
- **Marc Balle:** PhD at Universitat Pompeu Fabra

TEACHING

- **Lectures, seminars, and practical courses:** Responsible for the lecture of Image Registration in CAMP I & II; co-organizer of practical courses (Project Management and Software Development for Medical Applications and Computational Surgineering)

Research Interests

- Medical Image Processing
- Physics-based Deep Learning
- Generative Models
- Ultrasound Imaging

Languages

- Farsi (Native)
- English (Professional)
- German (Intermediate)

Publications

I have (first or last/co)-authored 3/5 journal articles, 7/14 conference papers, and 3/3 peer-reviewed workshop papers, and I am an inventor on a patent application in process. A detailed list is available below and on Google Scholar.

PEER-REVIEWED JOURNAL ARTICLES

- [J-8] A. Bitarafan, M. Mozafari, M. Azampour, M. Baghshah, N. Navab, et al. “Self-supervised 3D medical image segmentation by flow-guided mask propagation learning”. *Medical Image Analysis* 101 (2025).
- [J-7] M. Azampour, K. Mach, E. Fatemizadeh, B. Demiray, K. Westenfelder, et al. “Multitask weakly supervised generative network for MR-US registration”. *IEEE Transactions on Medical Imaging* 43.11 (2024).
- [J-6] M. Azampour, M. Tirindelli, J. Lameski, M. Gafencu, E. Tagliabue, et al. “Anatomy-aware computed tomography-to-ultrasound spine registration”. *Medical Physics* 51.3 (2024).
- [J-5] M. Gafencu, Y. Velikova, M. Saleh, T. Ungi, N. Navab, T. Wendler, and M. Azampour. “Shape completion in the dark: completing vertebrae morphology from 3D ultrasound”. *International Journal of Computer Assisted Radiology and Surgery* 19 (2024).
- [J-4] Y. Velikova, W. Simson, M. Azampour, P. Paprottka, and N. Navab. “Cactuss: Common anatomical CT-US space for US examinations”. *International Journal of Computer Assisted Radiology and Surgery* 19.5 (2024).
- [J-3] M. Ansari, Y. Yang, S. Balakrishnan, J. Abinahed, A. Al-Ansari, M. Warfa, M. Azampour, and et al. “A lightweight neural network with multiscale feature enhancement for liver CT segmentation”. *Scientific Reports* 12.1 (2022).

- [J-2] D. Grzech, M. Azampour, H. Qiu, B. Glocker, B. Kainz, and L. Folgoc. “Uncertainty quantification in non-rigid image registration via stochastic gradient Markov chain Monte Carlo”. *Melba Journal* 1 (2021).
- [J-1] A. Segato, C. D. Vece, S. Zucchelli, M. D. Marzo, T. Wendler, M. Azampour, et al. “Position-based dynamics simulator of brain deformations for path planning and intra-operative control in keyhole neurosurgery”. *IEEE Robotics and Automation Letters* 6.3 (2021).

PEER-REVIEWED CONFERENCE AND WORKSHOP PAPERS

- [C-27] F. Duelmer, M. Azampour, and N. Navab. “UltraScatter: Ray-based simulation of ultrasound scattering”. *2025 IEEE International Ultrasonics Symposium (IUS)*. 2025.
- [C-26] F. Duelmer, M. Azampour, M. Wysocki, and N. Navab. “UltraRay: Introducing full-path ray tracing in physics-based ultrasound simulation”. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2025.
- [C-25] M. Gafencu, Y. Velikova, N. Navab, and M. Azampour. “US-X Complete: A multi-modal approach to anatomical 3D shape recovery”. *International Workshop on Shape in Medical Imaging*. 2025.
- [C-24] J. Janelidze, L. Folle, N. Navab, and M. Azampour. “Tubular anatomy-aware 3D semantically conditioned image synthesis”. *MICCAI Workshop on Deep Generative Models*. 2025.
- [C-23] S. Joutard, M. Stollenga, M. B. Sanchez, M. Azampour, and R. Prevost. “HyperSORT: Self-organising robust training with hyper-networks”. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2025.
- [C-22] B. Møller, S. Amiri, C. Igel, K. Wickstrøm, R. Jenssen, M. Keicher, et al. “NEMt: Fast targeted explanations for medical image models via neural explanation masks”. *Northern Lights Deep Learning Conference*. 2025.
- [C-21] M. Sanchez, S. Joutard, M. Azampour, and R. Prevost. “Why is patch size important? Dealing with context variability in segmentation networks”. *Medical Imaging with Deep Learning – Short Papers*. 2025.
- [C-20] L. R. Vargas, M. Menten, J. Paetzold, N. Navab, and M. Azampour. “Skelite: Compact neural networks for efficient iterative skeletonization”. *International Conference on Information Processing in Medical Imaging (IPMI)*. 2025.
- [C-19] M. Wysocki, F. Duelmer, A. Bal, N. Navab, and M. Azampour. “UltrON: Ultrasound occupancy networks”. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2025.
- [C-18] V. Yesilkaynak, V. Duque, M. Wysocki, M. Azampour, N. Navab, et al. “UltraNBA: Neural bundle adjustment for pose refinement in 3D freehand ultrasound”. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 2025.
- [C-17] M. Domínguez, Y. Velikova, N. Navab, and M. Azampour. “Diffusion as sound propagation: Physics-inspired model for ultrasound image generation”. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2024.
- [C-16] F. Duelmer, W. Simson, M. Azampour, M. Wysocki, A. Karlas, and N. Navab. “Phocus: Physics-based deconvolution for ultrasound resolution enhancement”. *International Workshop on Advances in Simplifying Medical Ultrasound*. 2024.
- [C-15] M. Fehrentz, M. Azampour, R. Dorent, H. Rasheed, C. Galvin, A. Golby, et al. “Intraoperative registration by cross-modal inverse neural rendering”. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2024.
- [C-14] D. Grzech, L. Folgoc, M. Azampour, A. Vlontzos, B. Glocker, N. Navab, et al. “Unsupervised similarity learning for image registration with energy-based models”. *International Workshop on Biomedical Image Registration*. 2024.
- [C-13] Y. Velikova*, M. Azampour*, W. Simson, M. Esposito, and N. Navab. “Implicit neural representations for breathing-compensated volume reconstruction in robotic ultrasound”. *2024 IEEE International Conference on Robotics and Automation (ICRA)*. 2024.
- [C-12] Z. Zhang, B. Zhang, M. Azampour, S. Faghihroohi, A. Tomczak, et al. “Self-supervised vessel segmentation from X-ray images using digitally reconstructed radiographs”. *BVM Workshop*. 2024.

- [C-11] M. Azampour, Y. Velikova, E. Fatemizadeh, S. Dakua, and N. Navab. “Self-supervised probe pose regression via optimized ultrasound representations for US-CT fusion”. *International Conference on Medical Imaging and Computer-Aided Diagnosis*. 2023.
- [C-10] M. Mozafari, A. Bitarafan, M. Azampour, A. Farshad, et al. “VISA-FSS: A volume-informed self supervised approach for few-shot 3D segmentation”. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2023.
- [C-9] Y. Velikova, M. Azampour, W. Simson, V. G. Duque, and N. Navab. “Lotus: Learning to optimize task-based US representations”. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2023.
- [C-8] M. Wysocki*, M. Azampour*, C. Eilers, B. Busam, M. Salehi, and N. Navab. “Ultra-NeRF: Neural Radiance Fields for Ultrasound Imaging”. *Medical Imaging with Deep Learning (MIDL)*. 2023.
- [C-7] B. Zhang, S. Faghihroohi, M. Azampour, S. Liu, R. Ghotbi, H. Schunkert, et al. “A patient-specific self-supervised model for automatic X-Ray/CT registration”. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2023.
- [C-6] A. Bitarafan, M. Azampour, K. Bakhtari, M. S. Baghshah, et al. “Vol2Flow: Segment 3D volumes using a sequence of registration flows”. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2022.
- [C-5] D. Grzech, M. Azampour, B. Glocker, J. Schnabel, N. Navab, B. Kainz, et al. “A variational Bayesian method for similarity learning in non-rigid image registration”. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022.
- [C-4] B. Jian, M. Azampour, F. D. Benetti, J. Oberreuter, C. Bukas, A. Gersing, et al. “Weakly-supervised biomechanically-constrained CT/MRI registration of the spine”. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2022.
- [C-3] M. Tirindelli, C. Eilers, W. Simson, M. Paschali, M. Azampour, and N. Navab. “Rethinking ultrasound augmentation: A physics-inspired approach”. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2021.
- [C-2] H. Hase*, M. Azampour*, M. Tirindelli, M. Paschali, W. Simson, et al. “Ultrasound-guided robotic navigation with deep reinforcement learning”. *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2020.
- [C-1] M. Azampour, A. Ghaffari, A. Hamidinekoo, and E. Fatemizadeh. “Manifold learning based registration algorithms applied to multimodal images”. *2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. 2014.

PREPRINTS & MANUSCRIPTS

- [A-2] M. Gafencu, R. Shaban, Y. Velikova, M. Azampour, and N. Navab. *Shape completion and real-time visualization in robotic ultrasound spine acquisitions*. arXiv preprint arXiv:2508.08923. 2025.
- [A-1] M. Wysocki, M. Azampour, F. Tristram, B. Busam, and N. Navab. *Beyond Ultra-NeRF: Explainable neural fields for ultrasound*. Preprint. 2025.

Summary of Research Contributions

1. Physics-Aware Generative Ultrasound Modeling (Ray Tracing, Diffusion, NeRF). I introduced a family of physics-integrated generative models for ultrasound simulation, domain translation, and volumetric reconstruction. *UltraRay* provides fast, full-path acoustic ray tracing for realistic simulation; the *Physics-Inspired Diffusion (B-Maps)* scheduler incorporates depth-dependent noise to mimic ultrasound attenuation, yielding large gains in perceptual quality (e.g., Liver FID 73.4 $\rightarrow$ 0.86); and *UltraNeRF* is the first ultrasound-specific neural radiance field with physics-based volumetric rendering for view-dependent synthesis. Together, these works form a found-

dational stack for simulation, virtual re-scanning, and autonomous ultrasound systems.

References: Duclmer, Azampour et al., MICCAI 2025; Dominguez, Azampour et al., MICCAI 2024; Wysocki, Azampour et al., MIDL 2023.

2. Scalable Weakly Supervised MR–Ultrasound Registration. I developed *ProRegGAN*, the first multitask weakly supervised MR–US registration framework that requires no ultrasound annotations. A dual-path pseudo-US generator and a structure-preserving MaskNCE objective enable robust cross-modal alignment. The model achieves state-of-the-art prostate MR–US fusion on a 600-patient dataset and generalizes to external cohorts.

Reference: Azampour et al., *IEEE TMI* 2024.

3. Anatomy-Aware CT–Ultrasound Registration for Spinal Navigation. I introduced the first anatomy-aware CT–US registration framework for spinal imaging, combining a structural ultrasound representation (SUR) with an anatomically weighted similarity metric. The method achieves ~ 2.4 mm TRE on robotic spine ultrasound acquisitions and outperforms both classical and learning-based baselines.

Reference: Azampour et al., *Medical Physics* 2024.

4. 3D Shape Completion and Anatomical Reconstruction from Sparse Ultrasound. I developed methods for ultrasound-first 3D anatomical reconstruction capable of recovering complete vertebral geometry from fragmentary and occluded ultrasound. *Implicit Morphometry Networks* reconstruct smooth, anatomically consistent vertebrae from sparse ultrasound observations, while *US-X Complete* fuses sparse ultrasound with a single X-ray via a coarse-to-fine variational completion pipeline, yielding major improvements in CD, EMD, and F1 scores (all $p < 10^{-6}$). These works enable low-radiation, CT-free 3D spine reconstruction suitable for navigation and robotics.

References: Gafencu, Azampour et al., *IJCARS* 2024; Gafencu, Velikova, Navab, Azampour et al., *SMIMI* 2025.

5. Task-Driven Ultrasound Representations and Autonomous Robotic Guidance. I advanced ultrasound toward autonomy through cross-modal representation learning and ultrasound-guided control. *CACTUSS* introduced a CT-informed intermediate representation for robust aortic segmentation (88–90% DSC across devices). *LOTUS*, introduced subsequently, is a differentiable ultrasound renderer whose parameters are optimized end-to-end for segmentation, achieving $> 89\%$ DSC without ultrasound labels. Finally, *Ultrasound-Guided RL Navigation* demonstrated one of the first reinforcement-learning-driven robotic navigation systems using ultrasound alone, achieving $\sim 83\%$ success on unseen trials.

References: Velikova, Simson, Azampour et al., *IJCARS* 2024; Velikova, Azampour et al., *MICCAI* 2023; Hase, Azampour et al., *IROS* 2020.